

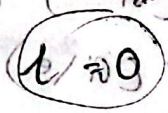
# Quiz - 3

E.P.A. 00000

(1) Every  $u-v$  walk contains a  $u-v$

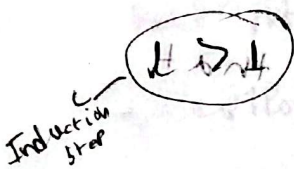
path

+ (0-2) Basic step



No edge  $\therefore (u=v) + (u=v) =$

This vertex is a  $u-v$  path of length 0



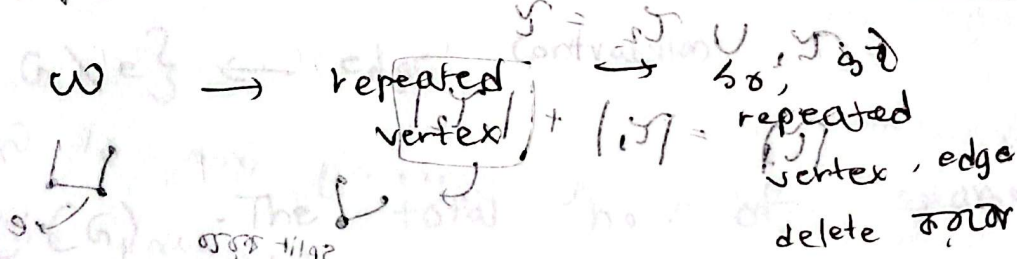
Let  $W$  be a  $u-v$  walk of length  $l$

$$|A| + |A| = |A \cup A|$$

$$|A \cap A| \rightarrow \text{no repeated vertex}$$

$$0 - |A| + |A| =$$

no repeated vertex  $\rightarrow$  path



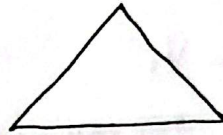
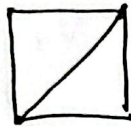
$(u/v) \gamma = |\gamma|$   $\downarrow$   $W'$  (shorter)

So,  $W'$  contained still  $+ |W'| = \gamma$

$$(u/v) \gamma + |W'| =$$

$$(u/v) \gamma + (u-v) \gamma =$$

(2) Every graph with  $n$  vertices and  $m$  edges has at least  $n - m$  components.



So, components are

$$n - k \geq n - m$$

$$k \leq m$$

We know, (edges)  $\rightarrow$  complete graph

Adding Edge  $\rightarrow$  Decreases the number of components by 0 or 1. / at most 1

An  $n$ -vertex graph with no edge has  $n$  components

So, we add  $m$  edges.  $\rightarrow$  can decrease the number of components at most  $m$ .

So, components  $= n - m$

$$n - m$$

$$\text{So, } k \geq n - m$$

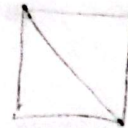
(3) Let  $G$  be a simple

$$n-k \leq m \leq \frac{(n-k)(n-k+1)}{2}$$

we know

$$n-k \geq n-m$$

$$m \geq n-k$$



→ Complete Graph (Maximum edges) no. of

Adding Edge  $n_i > n_j$  →  $H_i, H_j$  →  $n_i > n_j$

Gets replaced by

$$(n_i+1), (n_j-1)$$

Complete Graph → [Increase in the no. of edges]

$$(n_i+1)c_2 \text{ \& } (n_j-1)c_2$$

Increase in edges.

$$\rightarrow [(n_i+1)c_2 - n_i c_2] + [(n_j+1)c_2 - n_j c_2]$$

$$m - n \leq k \quad \text{or} \quad m - n$$

$$= n_{j+1}$$

$$= n_j - n_j + 1$$

$$= n - k + 1 \quad C_2 = \frac{(n-k)(n-k+1)}{2}$$

A Graph  $G$  is bi-partite if and only if

Let,  $x$

Part 1 2 partite sets  $(V_1 \& V_2)$

$x_1, x_2, x_3, \dots$  such that  $x_i \rightarrow x_{i+1}$  cycle of length 2 in  $G$

$x_1 \in V_1$

$x_2 \in V_2$

$x_3 \in V_1$

$x_4 \in V_2$

So, cycle is even cycle

Part 2

$G$  is Bipartite if and only if

every component is bipartite

$G \rightarrow$  connected bipartite

if

$x \rightarrow$  arbitrary vertex of  $G$

We construct  $V_1$  such that  $1 + iN - jN =$

$$V_1 = \{ y \mid d(x, y) \leq \frac{1 + iN - jN}{2} \}$$

Then  $V_2 = V - V_1 \rightarrow V_1$  &  $V_2$  independent sets

and expand to all

Trees &  $(v)$  at  $2$  or  $3$

(1) Every tree

more vertices  $v$  has at least 2 leaves

$P \rightarrow$  maximal path.

Let  $v$  on end vertices be  $u \in V$  &  $v \in X$

maximal path  $u \rightarrow v$  if  $u \in V$  &  $v \in X$

new degree:  $\rightarrow$  but if all  $v \in X$  the tree



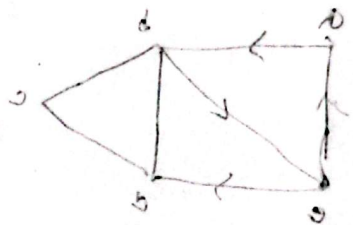
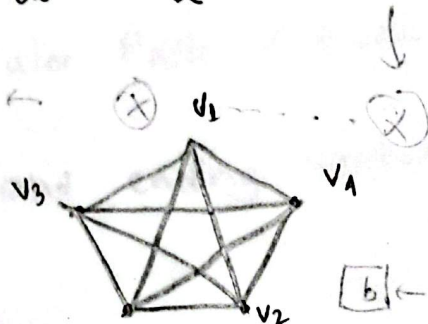
Two leaves of the tree

gives maximal path  $u \rightarrow v$  or more leaf

Q A connected graph is Euler graph if and only if each of its vertices

is of even degree

→ Let  $G$  be connected and also if  $G$  is an Euler graph. Let  $C$  be an Eulerian circuit of  $G$  with origin and terminates at vertex  $u$ .



If  $v$  is an internal vertex then  $d(v) = 2 \times$  number of times  $v$  occurs inside the Euler circuit

$d(v) = 2 \times$  number of times  $v$  occurs inside the Euler circuit

Path :- / Circuit  $u \rightarrow v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v_2 \rightarrow u \rightarrow v_2 \rightarrow v_1 \rightarrow v_4 \rightarrow v_1 \rightarrow u$

Euler circuit starts and ends at  $u$ .

$$\Rightarrow d(u) = \textcircled{2} + 2 \times \left\{ \begin{array}{l} \text{number of times} \\ u \text{ occurs inside } C \end{array} \right.$$

first & last

$\therefore$   $G$  has all the vertices of even degree

$\Rightarrow$  Conversely assume that each of the vertices of  $G$  has an even degree

we need to prove  $G$  is an Euler graph

Suppose  $G$  is not an Euler graph

then  $G$  has no Euler circuit.  
But each vertex of  $G$  has even degree.

$\hookrightarrow G$  contains Euler circuit

$\square$  Can be decomposed into circuits / has cycle decomposition

$\Rightarrow$  Let  $G$  be an connected graph  
it has closed walk which contains all edges of a graph

Degree of each vertex  $\rightarrow$  even

Let  $v_1$  be any vertex. So,  $v_1$ 's degree is even. So, at least 2 edges incident on it.

Let one of the edge between  $v_1$  &  $v_2$

Also  $v_2 \rightarrow$  even. Another edge incident.

Let the edge be  $v_2$  &  $v_3$

We continue in this way until no edge left.

Thus Graph  $G$  decomposed in a circuit.

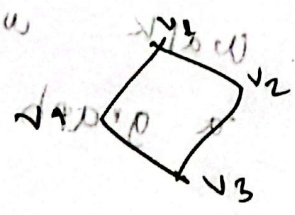
## Part II

Decomposed into a circuit

Euler graph

Let  $G$  be the union of edge disjoint circuit

Since in a circuit degree of each vertex is even



Euler graph

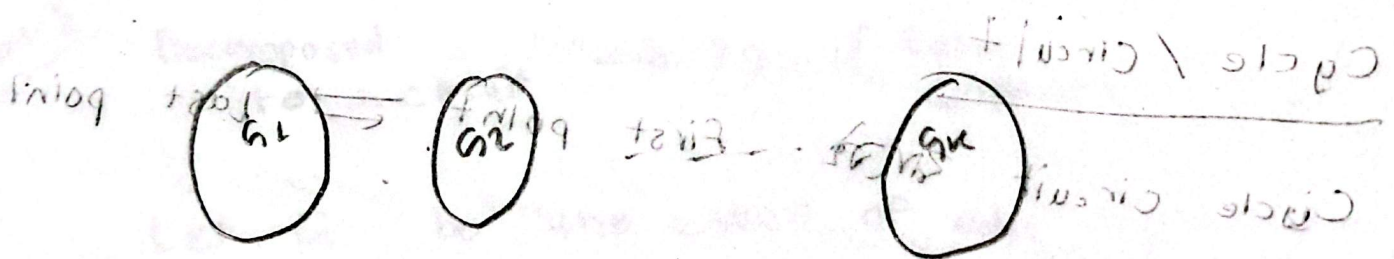
$\hookrightarrow$  If  $G$  is a hamiltonian then for every non-empty subset  $S \subseteq V(G)$  the number of connected components in  $G-S$  is at most  $|S|$ .

$\hookrightarrow G$  is hamiltonian,  $S \neq \emptyset$ ,  $S \subseteq V(G)$  and  $S$  is not empty.

let  $w \in S$  and  $C$  is hamiltonian cycle:

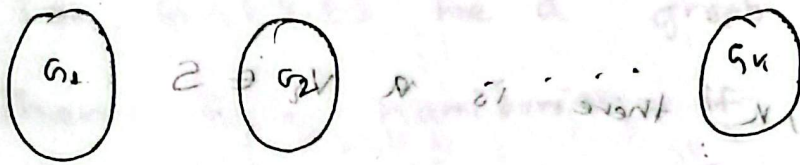
$w \xrightarrow{C} w$  cycle of  $G$

Components  $(G-S) = k$  and let  $G_1, G_2, \dots, G_k$  be the connected components of  $G-S$

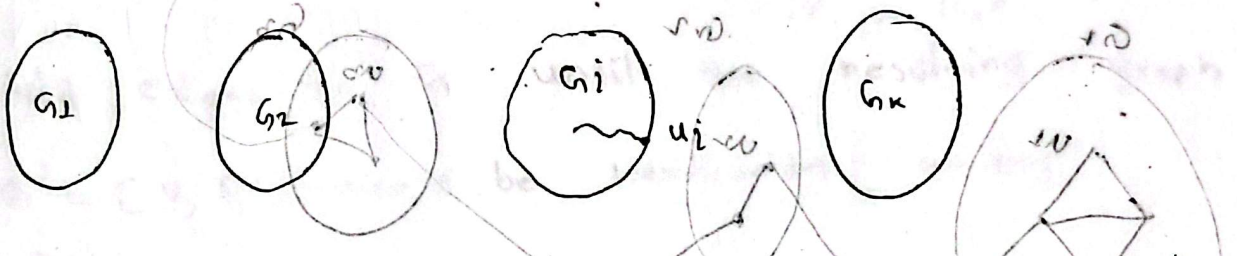


so, if  $n > 1$ , we give an direction



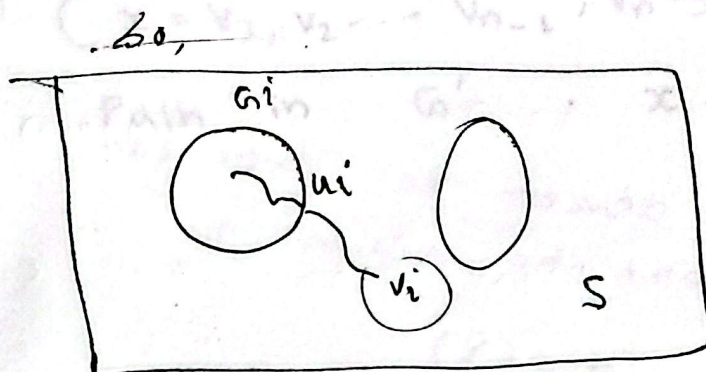


Let  $u_i$  be the last vertex of  $C_i$  that belongs to the component  $G_i$ . Let  $v_i$  be the next vertex of  $C$  after  $u_i$ .



$v_i \notin V(G_i)$  -  $v_i$  will belong to some other component. So, belong to  $G_j$ . There must be an edge connected by  $G_j$ . But that can't happen because it means  $\rightarrow$  components are connected.

$v_i \in V(G_j) \leftarrow$  mutually disjoint

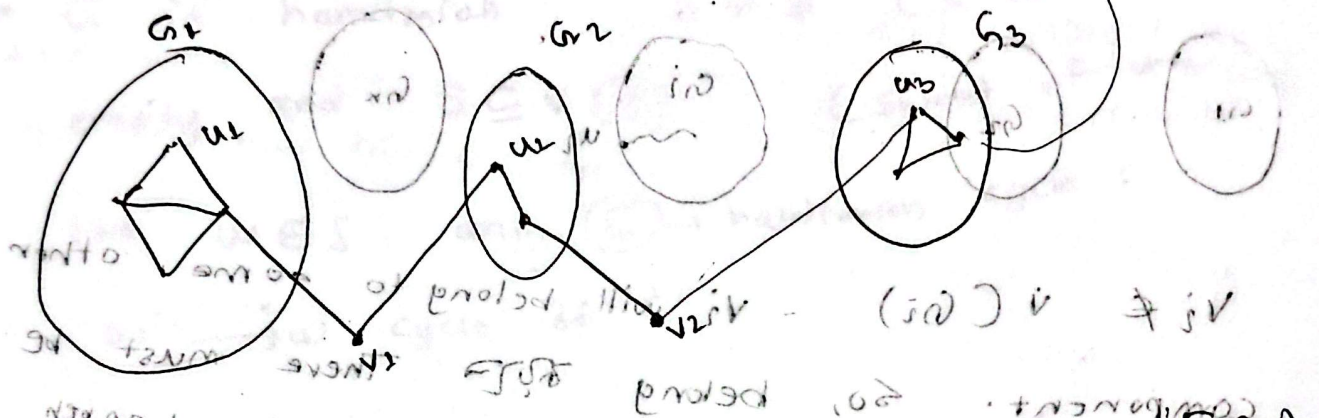


$v_i \neq v_j$

$\rightarrow v_i \in S$

For each  $i = 1, \dots, k$  there is a  $v_i \in S$

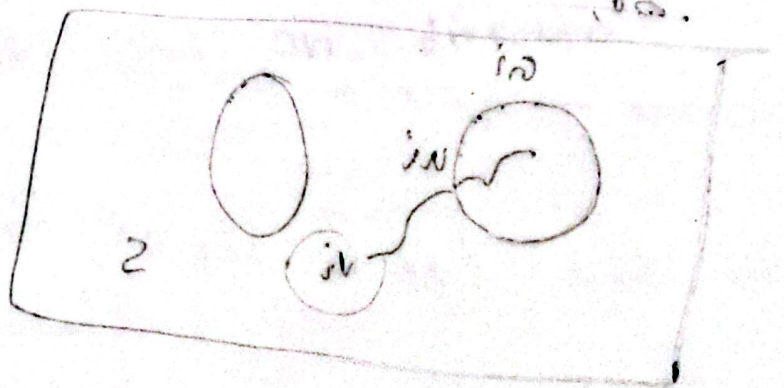
$|S| \geq k$  to select the last vertex of  $C$  let  $v$  be  $k = \text{Components}(G - S)$  belongs to the next vertex of  $C$  after  $v$



because  $C$  is a cycle and  $u_i, v_i \in E(G)$  for all  $i$

because it means  $v_i \in V(C)$

$v_i \neq v_j$   
 $v_i \in S$



Let  $G=(V,E)$  be a graph with  $n \geq 3$ .

Then  $G$  is hamiltonian if  $d_G(u) + d_G(v) \geq n$  for every pair of non-adjacent vertices  $u$  and  $v$  in  $G$ .

→ Contradiction → Let say  $G$  is not hamiltonian

Add edges to  $G$  until the resulting graph  $G'=(V,E')$  would be hamiltonian w. any additional edge.

So,  $x, y \in V$  such that  $x, y \notin E'$  so,  $x$  &  $y$  are not an element of edge set

not adjacent  $\rightarrow d(x) + d(y) \geq n$

Take  $x, y \in V$  such that  $x, y \notin E'$

$(x=v_1, v_2, \dots, v_{n-1}, v_n=y)$  is a Hamiltonian

Path in  $G'$   $x \rightarrow \dots \rightarrow y \rightarrow x$  → hamiltonian cycle

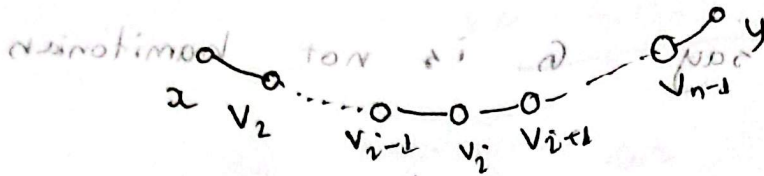
$$d(x) + d(y) \geq n - 1 - d(x)$$

can't add edge any more  
Degree sum for vertex

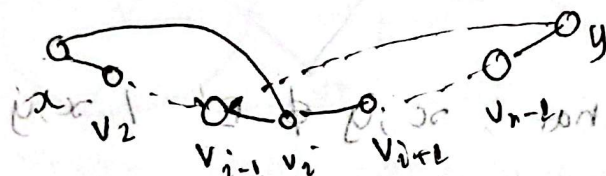
Let  $G = (V, E)$  be a graph with  $n \geq 3$  and  $\deg(v) \geq 2$  for every  $v \in V$ .

For all  $v_i$  where  $2 \leq i \leq n$ ,  $x_{v_i} \in E$ ,  $y_{v_{i-1}} \notin E$ .

$$2 \leq i \leq n$$



Consider the path  $P = (v_2, v_{i-1}, v_i, v_{i+1}, \dots, v_{n-1})$ . The vertices  $v_2$  and  $v_{n-1}$  are adjacent to vertices outside the path.



otherwise

$(x, v_i, v_{i+1}, y)$

Let  $G' = (V, E')$  be a graph with  $n \geq 3$  and  $\deg(v) \geq 2$  for every  $v \in V$ .

Let  $G'$  be a Hamiltonian cycle in  $G'$ .

Let  $x = v_1, v_2, \dots, v_{n-1}, v_n = x$ .

$$\deg(y) \leq n-1$$

for every vertex  $x$  is adjacent to  $y$ , there can be a vertex  $y$  that can't be adjacent to  $x$ .

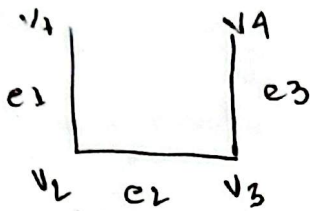
$$\deg(y) \leq n-1 - \deg(x)$$

$$\deg(y) + \deg(x) \gg \leq n-1$$

which is not true. so,  $G$  is hamiltonian

## Trees

Every tree with two or more vertices has at least two leaves.



$$d(v_2) = 2 \quad d(v_1) = 1$$

$$d(v_3) = 2 \quad d(v_4) = 1$$

$v_1$  &  $v_4$  are end vertices / leaves.

Tree  $n$  vertices  $\rightarrow (n-1)$  edges

$(n-1)$  edges  $2(n-1)$  degrees  $= 2(n-1)$  which equals to  $(2n-2)$  degree

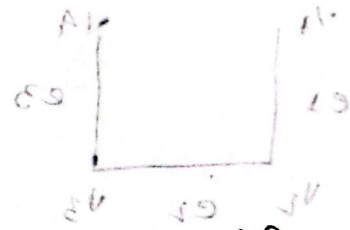
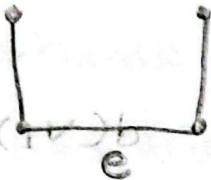
Since connected graph / tree have no isolated vertices all vertices must have minimum degree 1.

$$\text{So, } (2n-2) - n = (n-2) \text{ remaining degrees.}$$

So, if  $(n-2)$  degree distributes on  $(n-2)$  vertices. the two vertices become leaves.

Hence, there must be at least 2 leaves.

Every edge in a tree is a cut edge.



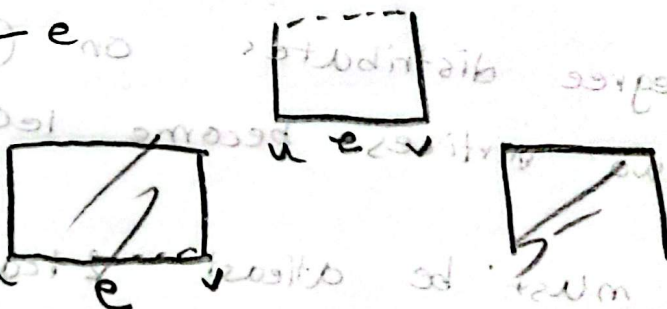
cut edge  $\rightarrow$  is an edge that if

removed splits a connected graph into 2 disconnected parts.

Let  $T$  be Tree

The edge  $e = uv$

By contradiction, let  $e$  not be a cut edge. So, we can delete  $e$  and still have a connected part.



Not possible cause there exist a cycle. So, every edge is a cut edge.

▢ Forest with  $n$  vertices and  $k$  components has  $n - k$  edges.

Forest  $F$  with  $n$ ,  $k$  components

$k$  components  $\rightarrow T_1, T_2, T_3, \dots, T_k$  ( $k$  Trees)

Tree  $T_i$ ,  $n_i$  vertices  $\rightarrow n_i - 1$  edge

$$\begin{aligned} k \text{ Trees, } m &= \sum_{i=1}^k (n_i - 1) \\ &= \sum_{i=1}^k n_i - \sum_{i=1}^k 1 \\ &= n - k \end{aligned}$$

▢ Forest with  $n$  vertices and  $m$  edges contain  $n - m$  components.

Tree we know  $m = n - k$

So,  $k = n - m$

▢  $n$  vertices,  $m$  edges, and  $k$  components graph will be acyclic

Base step : we know graph  $\rightarrow n = n - 1$   
 $m = n - 1$

Graph  $\rightarrow$   $n$  vertices  
 $m$  edges  
 $n-m$  components

Let  $k$  be components  $\rightarrow k = n - m$

Let the connected components of the graph be  $k_1, k_2, \dots, k_k$

$$\sum_{i=1}^k n_i = n, \quad \sum_{i=1}^k m_i = m$$

Graph

A connected graph with  $v$  vertices

must have  $v-1$  edges to remain connected.

$\rightarrow$  Exactly  $v-1$  edge  $\rightarrow$  No cycle

More than  $v-1$  edge  $\rightarrow$  At least one cycle

For each component  $k_i$ , we know

$$m_i \geq n_i - 1$$

$$\sum_{i=1}^c m_i > \sum_{i=1}^c (n_i - 1)$$

$$m \geq n - 1$$

$$m \geq n - k$$

A.T.A graph has exactly  $n - m$  components.

$$m = n - k$$

$$m = n - 1$$

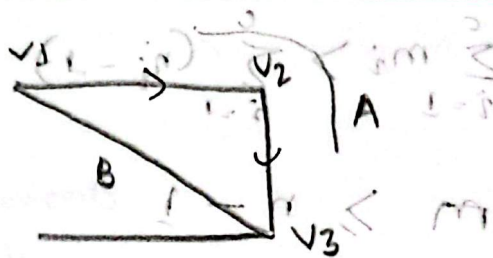
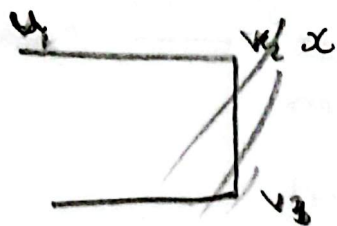
True for every connected component

→ Tree

A Simple graph  $G$  is a tree if and only if for each pair of vertices  $u$  and  $v$  there is a unique path  $(u-v)$  path in  $G$ .

Contradiction :-

Let  $u_1$  &  $v_3$  connected by two different paths. So, the union of these two paths will contain a circuit.



B is the union of two paths.  $m$   
 path  $\cdot A$  is also a path. Union of  
 $A \& B \rightarrow$  loop  $\rightarrow$  If loop can't  
 exist  $\rightarrow$  so, there exist 1 path only/unique.

$\Rightarrow G$  is tree

path between every possible pair

of vertices  $\rightarrow$  connected + 1 unique

path  $\rightarrow$  no loop  $\rightarrow G$  is tree

**Spanning Tree :-**

A connected Sub-Graph  $S'$  of a

graph  $G(V, E)$  is said to be  
 spanning if

- 'S' should contain all vertices of 'G' / ~~cannot~~ vertex isolated ~~over over~~
- 'S' should contain  $(|V| - 1)$  edges
- No cycles

□ Every connected graph contains a spanning tree.

(v) Let  $G$  be a connected graph

(T) <sup>minimal connected</sup> → spanning subgraph of  $G$

$$|V(T)| = |V(G)|$$

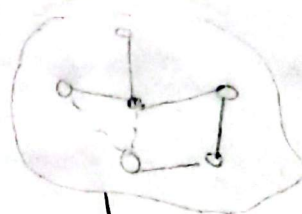
→ T has minimum number of edges among all connected spanning subgraph of  $G$

→ So, for any edge  $e$ ,  $T - e \rightarrow$  disconnected

Every edge is cut edge in  $T$

So,  $T$  cannot contain a cycle.  $T \rightarrow$  acycle.

$T$  (acyclic + connected graph)  $\rightarrow T$  is tree.

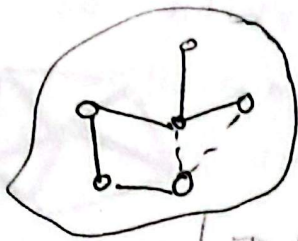
$$\begin{aligned}
 \tau_1 \cap \tau_2 &= \emptyset \\
 \tau_1 \cup \tau_2 &= \tau \\
 |\tau| &= |\tau_1| + |\tau_2|
 \end{aligned}$$


### Lemma 4.4.2

Let  $G = (V, E)$  be a connected graph and  $S$  is a subset of  $E$  such that  $G \setminus S$  is a disconnected graph, then every spanning tree of  $G$  has an edge in  $S$ .

By contradiction,

Proof: Let  $T$  is a spanning tree of  $G$ . Assuming  $T$  does not contain any edge that belongs to  $S$ .



$$G \quad |T| + |T| = |T|$$

$$\phi = \{T\} \quad S = \{e_1, e_2, \dots, e_n\}$$

$$T = T \cup T \quad T = \{e_{q+1}, e_{q+2}, \dots, e_n\}$$

Since, after removing all the  $e \in S$  still the graph is connected and acyclic

so it still stays a spanning tree

but it is not possible to form a spanning tree

question / statement.

So, a contradiction is proved every spanning tree of  $G$  has at least one edge in  $S$ .

$\{G \setminus e\} \leftarrow$  edge contraction

$\mathcal{T}(G)$ . The total no. of spanning trees in a connected graph

# Lemma 4.4.3

Let  $G$  be a connected graph and

let  $e$  be an edge then prove (1)

$$\tau(G)$$

$$= \tau(G - e) + \tau(G \setminus e)$$

$$= \tau(G)$$

The no. of those spanning trees that

do not contain  $e$

is the no. of those spanning trees that contain  $e$ .

$$\tau_1 = \tau(G - e)$$

$$\tau_1 \cap \tau_2 = \emptyset$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$\tau_1 \cup \tau_2 = \tau$$

$$|\tau| = |\tau_1| + |\tau_2|$$

split into

(vertex)

$$|\tau_2| = \tau(G \setminus e)$$

$$\tau = |\tau_1| + |\tau_2|$$

$$= |\tau_1| + \tau(G \setminus e)$$

$$= \tau(G - e) + \tau(G \setminus e)$$

Lemma: Let  $G$  be a graph with  $n$  vertices, then any 2 of the following statements imply the 3rd.

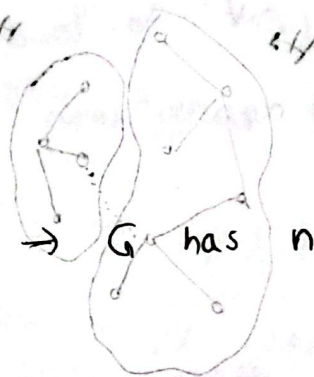
- (i)  $G$  is connected
- (ii)  $G$  contains no cycles
- (iii)  $G$  has  $n-1$  edges

i & ii  $\rightarrow$  iii

i & iii implies ii

i & ii  $\rightarrow$  (iii)

$G$  is connected and acyclic



Method: Strong Induction

$$P(1) \rightarrow P(n-1) \Rightarrow P(n)$$

Basis Step:

for  $n=1$ ,  $|E(G)| = 0 = 1-1 = 0$

$P(1)$  is true.

Inductive Step:  $\rightarrow P(1), P(2), \dots, P(n-1)$  are true.

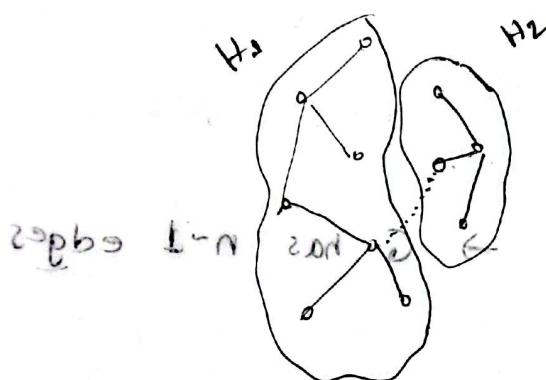
Let the statement  $P(n)$  be true for  $n=1, 2, 3, \dots, n-1$

We'll prove that  $P(n)$  is also true.

$$1-1 \rightarrow 1-(n-1) = 0$$

Let  $G$  has  $p$  vertices, Since  $G$  is connected and acyclic.  $\therefore G$  is a tree. by lemma 4.2.2, every edge in  $G$  is a cut edge.

$\therefore$  If we delete an edge  $e$  from  $G$ , we have two components  $H_1$  &  $H_2$  where both  $H_1$  &  $H_2$  are connected and acyclic.



Now,  $P(n-1) \Rightarrow P(n)$

$P(1) \dots P(n-1) \Rightarrow P(n)$

$n_1 = \text{No. of nodes in } H_1 < p$   
 $n_2 = \text{No. of nodes in } H_2 < p$

Now according to the hypothesis,

$$0 = 1 - 1 = |E(H_1)| = n_1 - 1 + |E(H_2)| = n_2 - 1$$

Here,

$$n_1 + (n_2 - 1) = p$$

$$|E(G)| = |E(H_1)| + |E(H_2)| + 1$$

$$n_1 - 1 + n_2 - 1 + 1 = p - 1$$

$$\rightarrow (n_1 + n_2) - 1 \rightarrow p - 1$$

(i) & (iii)  $\rightarrow$  (ii)

If  $G$  is connected and has  $n-1$  edges then  $G$  is acyclic.

Method of Contradiction

Proof: Let  $G$  has  $k$  cycles. Now we start deleting one edge from each of the cycles until the graph is acyclic.

Since from 3.1.4: No edge in a cycle is a cut edge. After removal of  $k$  edges from  $k$  cycles, the new graph  $G'$  remains connected.

Here,

$G'$  is a connected and acyclic graph

$\therefore$  from (i) & (ii)  $\Rightarrow$  (iii),  $G'$  has  $n-1$  edges

But after deletion of  $k$  edges,  $G'$

But,  $G$  had  $n-1$  edges  $\rightarrow$  still graph  
 $\therefore$  we couldn't delete any edge.

$\Rightarrow G$  had no cycle.

$(ii) \rightarrow (iii) \& (i)$   
 $ii \& iii \Rightarrow$  If  $G$  has no cycle and is connected then  $G$  is a tree and has  $n-1$  edges.

Contradiction: Let  $G$  be disconnected. Let  $H_1, H_2, \dots, H_k$  be the components of  $G$ . Each  $H_i$  is connected and acyclic by (ii).

From (i) & (ii)  $\Rightarrow$  (iii)

$H_i$  has  $n_i - 1$  edges  
 $\forall i \in \{1, 2, \dots, k\}$

Here,  $\sum_{i=1}^k n_i = n$

$$|V(H_i)| = n_i$$

$$\sum_{i=1}^k |E(H_i)|$$

$$= \sum_{i=1}^k (n_i - 1)$$

$$\left( \sum_{i=1}^k n_i \right) - k$$

Substituting  $\sum_{i=1}^k n_i = n \dots \dots \dots (1)$

$$\sum_{i=1}^k |E(H_i)| = \sum_{i=1}^k |E(H_i)| = n - k \quad \left. \begin{array}{l} n - k = n - 1 \\ k = 1 \end{array} \right\} \begin{array}{l} G \text{ was connected.} \end{array}$$

Corollary  $\rightarrow$  4.2.4

A Forest with  $n$  vertices and  $k$  components then what will be the number of edges

(2) A Forest with  $n$  vertices and  $m$  edges contain how many components:

(3) Graph with  $n$  vertices,  $m$  edges  $m-1$  components is acyclic

Lemma 4.2.2.5

## Quiz - 4

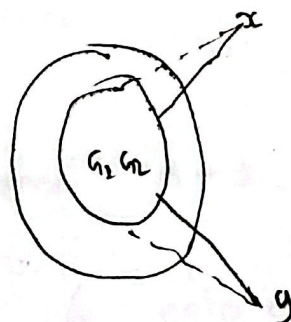
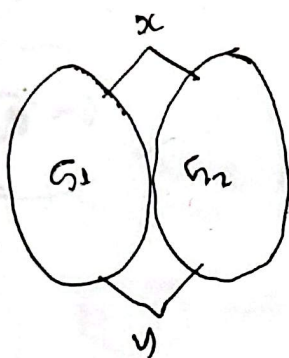
2-connected planar graph is unique if and only if  $G$  is a subdivision of a 3-connected graph / Every 3-connected graph has a unique plane embedding

$\Rightarrow$  We assume  $G$  is a 2-connected planar graph not a subdivision of a 3-connected

Not a subdivision  $\rightarrow$  separate pair of vertices  $\{x, y\}$

Split  $G$  at these 2 vertices  $\rightarrow$  2 subgraphs

$G_1$  &  $G_2$  .  $G_1, G_2$  not simple path.



(new embedding after flipping)

Therefore plane embedding of  $G$  is not unique.  
on subdivision

Q Let  $G$  be a connected plane graph...

$$n, m, f \rightarrow n - m + f = 2$$

$$f = 2 + m - n$$

Base case  $\rightarrow m = 0$  or  $1$

Assume  $m \geq 2$  ( $f \geq 1$ )

$G \rightarrow$  connected plane graph

(i)  $G$  is a tree

As it is a tree, it must have one vertex with 1 degree (leaf). Removing

this vertex  $\rightarrow G - v \rightarrow$  now connected  
 $m - 1$  edges

$G - v$  has  $n - 1$  vertices  
 $m - 1$  edges

(ii) ...  $f$  (tree have no enclosed cycles)  
 $f = 1$

So,

$$(n-1) - (m-1) + f = 2 \rightarrow m = n - 1$$

(ii)  $G$  not tree

Not Tree  $\rightarrow$  so  $G$  cycle. So, cycle of edge for  $m$  faces.

$G - e$

$$G - e \rightarrow \begin{matrix} n \\ m \end{matrix} \rightarrow \begin{matrix} n \\ m-1 \end{matrix}$$

$f - 1$  faces

$$G \text{ is connected} \Rightarrow n - (m-1) + (f-1) = 2$$

$$\downarrow$$

$$n - m + f = 2$$

$G$  is planar ( $n \geq 3$ ) and  $m \leq 3n - 6$   
 equality holds if  $G$  is maximal planar

Every edge in a planar graph borders at most 2 faces.  
 $\sum d(F) = 2m$

$G$  is simple graph  $\rightarrow$  no loops or with  $n \geq 3$   
 $\rightarrow d(F) \geq 3$  for every face

(i) & (ii)  $\rightarrow 2m \geq 3f$

$f \leq \frac{2m}{3}$

We know,

$$f = 2 - n + m$$

$$2 - n + m \leq \frac{2m}{3}$$

Every maximal planar graph has  $m = 3n - 6$  edges.   
 or  $m \leq 3n - 6$  for planar graphs with  $n \geq 3$  vertices.

Maximal planar graph  $\rightarrow$  + edge  $\rightarrow$  planarity is lost

Every face is exactly a triangle (Triangulation)

$$2m = 3f \rightarrow f = \frac{2m}{3}$$

(multiply by 2)  $\rightarrow$  substituting  $m = 3n - 6$

$$m = 3n - 6$$

In planar graph, vertex of degree  $\leq 5$  most

$\rightarrow$  we know

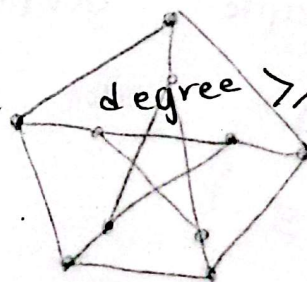
$$m \leq 3n - 6 \quad [ \text{if } G \text{ is planar graph} ]$$

Contradiction

All vertices

have

degree  $\geq 6$



$$\downarrow$$

$$m \geq 3n$$

$\rightarrow$  combining

$$3n \leq m \leq 3n - 6 \rightarrow 3n \leq 3n - 6$$

$$0 \leq -6 \quad [ \text{Impossible} ]$$

Every maximal planar graph has  $\geq 4$  vertices at least 4 vertices

Maximal planar graph  $\rightarrow$  edge set  $\rightarrow$  edge set

So,  $G$  maximal planar with  $n \geq 4$

and  $m$  edges  $\rightarrow 2 = \sum \deg(v) = 2m$

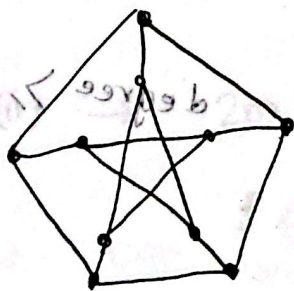
$$m = 3n - 6 \quad \text{for maximal planar}$$

We know,

$$\sum \deg(v) = 2m$$

$$\sum \deg(v) = 2(3n - 6) = 6n - 12$$

Petersen Graph



vertices

edges

$$n \leq m$$

$$n \geq m$$

$$n \geq m \geq n$$

$$n \geq m \geq n$$

Ⓟ Bipartite graph properties

$$m \leq 2n - 4 \quad (1 + \Delta) \quad \Delta \text{ is odd} \quad n \geq 3$$

Bipartite Graph

→ Simple connected graph so

→ no cycles of odd length

(No  $C_3, C_5$ )

→ smallest possible cycle 4 edges →  $d(F) \geq 4$   
 ↓  
 degree of every face is at least 4

$$\sum d(F) = 2m$$

$$\rightarrow 2m \geq 4f$$

$$f \leq \frac{m}{2}$$

$$1 + \Delta \geq \chi$$

We know,

$$m + f = 2$$

$$f = 2 - n + m$$

$$2 - n + m = f$$

$$2 - n + m \leq \frac{m}{2}$$

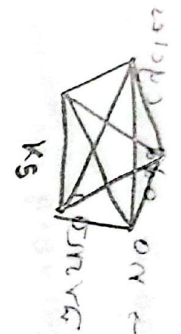
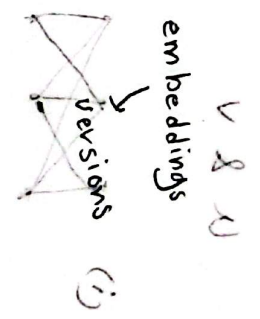
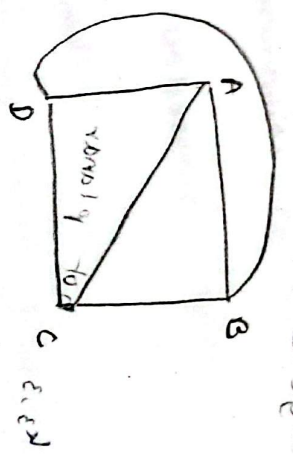
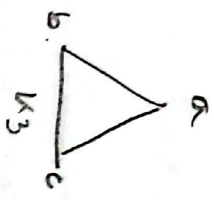
$$4 - 2n + 2m \leq m$$

$$4 - 2n + m \leq 0$$

$$m \leq 2n - 4$$

# Planar Graphs

Graph is called planar graph if it can be drawn in a plane (2D) in a such way that no edges cross over



$K_5 \rightarrow K_5$  non planar graph

$K_5 \rightarrow K_{3,3}$  bipartite graph

$K_5 \rightarrow K_{3,3}$  bipartite graph

## Kuratowski's Theorem

A graph is planar if and only if it contains no subgraph that is a subdivision of either  $K_5$  or  $K_{3,3}$

$K_5$  or  $K_{3,3} \rightarrow$  non planar

Every subgraph of planar graph  $\rightarrow$  planar



subdivision of non-planar graph  $\rightarrow$  non planar

2/10/2017

L3 → Let  $G$  be a 2-connected graph

and  $u, v$  are vertices of  $G$ . There exists a cycle in  $G$  that includes both  $u$  and  $v$ .

$u$  &  $v$

connected



$K_{3,3}$

not planar

But contradiction of Kuratowski's theorem.  $K_{3,3}$  is planar.

$n = 6$ , and  $G$  is bipartite → no odd cycles  
edges. we know, at least there are 4 edges

$$50, f \leq 20/4 \rightarrow 4.5$$

$$50, v - e + f = 1.5 \rightarrow 1.5 \neq 2.0$$

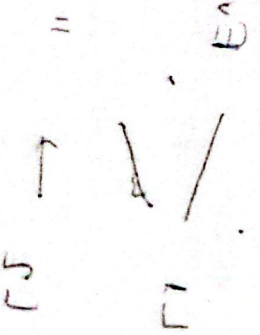
So,  $K_{3,3}$  is not planar

moreover, it is not planar

i)  $K_5$  is not planar. Also,  $K_5$  is the complete graph on 5 vertices.  $K_5 \rightarrow v = 5, e = 10$ . It is not bipartite.

we can show  $K_5$  is not planar by using the fact that it has 5 vertices and 10 edges.

non-bipartite graph is not planar.



Every simple graph of the max degree  $\Delta$  has a  $(\Delta+1)$ -coloring.

Let the vertices of the graph be:-

$v_1, v_2, v_3, \dots, v_n$

Now coloring  $v_i$   
 (i) At most  $\Delta$  neighbours. So,  $\Delta$  are colors for bidden

So,  $\Delta+1$  guarantees one available color  
 $\chi(G) \leq \Delta+1$

$G \rightarrow$  a connected simple graph

$G \rightarrow$  not an odd cycle

$G \rightarrow$  not a complete graph

$\Delta \neq (n-1)$

$\chi(G) \leq \Delta+1$

$\Delta \geq (n-1)$

$\chi(G) \leq \Delta+1$

Every simple graph of the maximum degree  $\Delta$  which is not complete graph has  $\Delta$ -coloring. /  $\chi \leq \Delta$

For complete graph

$$\chi(K_n) = n$$

Degree of each vertex  $\rightarrow \Delta = n-1$

$$\Delta = n-1 \rightarrow \Delta + 1 = n$$

So, chromatic number is

$$\chi(K_n) = \Delta + 1$$

as it requires more than  $\Delta$  colors.

So,

$G \rightarrow$  a connected simple graph

$G \rightarrow$  is not an odd cycle

$G \rightarrow$  not a complete graph

$$\text{So, } \chi(K_n) \neq \Delta$$

$\chi(G) = k \rightarrow k$ -chromatic /  $k$ -coloring

So,

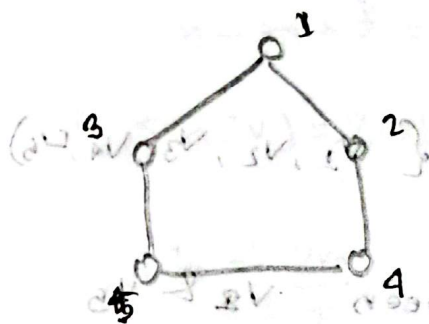
$\chi(G) \leq \chi(K_n) \leq \Delta$  for not a complete graph

$\rightarrow \chi(K_n)$  has a  $\Delta$  coloring

Every simple planar graph is 6-colorable

Basis step :  $P(n \leq 5) \rightarrow$  Graph can be colored.

(Every planar graph contains a vertex with degree  $\deg(v) \leq 5$ )

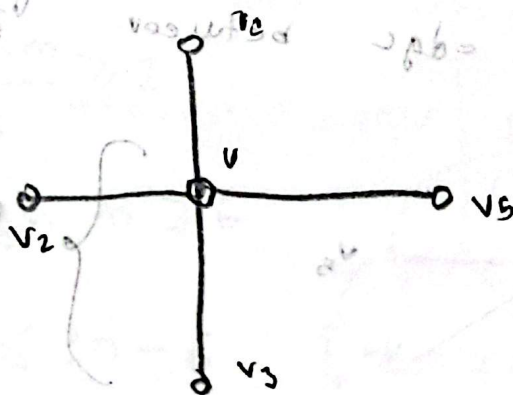


(i) Induction step  $\rightarrow$  Let  $k=4$

$\deg(v) \leq 4 \rightarrow$  True

(ii) Verify for  $k+1$

$\deg(v) \leq 5$

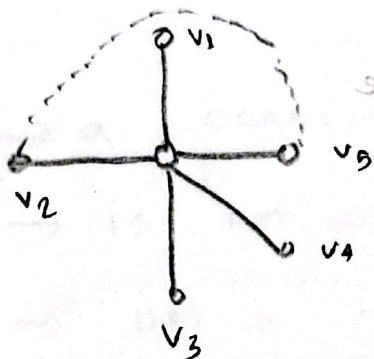


Every simple planar graph is 3-colorable  
 Basis step:  $6 (0 \leq 2) \rightarrow$  graph is 3-colorable  
 (Every planar graph is 3-colorable)  
 Adding a vertex  $v$  with  $\deg(v) \leq 2$

Adding  $\therefore$  Cases for (iii)

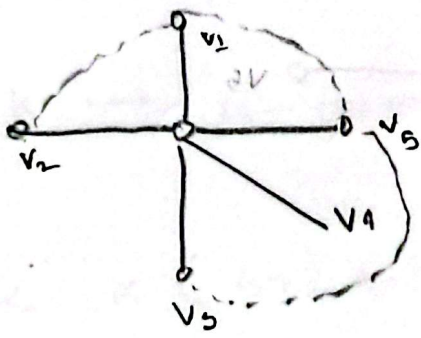
No direct edge between  $\{v_1, v_2, v_3, v_4, v_5\}$

Assuming direct edge between  $v_2$  &  $v_5$



$\deg(v) \leq 5$  colors  
 (iii) Verified for  $K+1$   
 $\deg(v) \geq 2$

Assuming direct edge between  $v_3$  &  $v_5$



$\deg(v) \leq 5$  colors

Every simple planar graph is 6-colorable

$$\frac{d_v}{v} < 6 \quad / \quad \frac{dn}{n} < 6$$

We know,

$$n - m + f = 2$$

$$\sum d(v) = 2m$$

(Handshaking Lemma)

$\sum d(v) \leftarrow$  sum of all degree

$$k = \frac{2m}{f}$$

$$f \leq \frac{2m}{3}$$

$$n - m + f = 2$$

$$n - m + \frac{2m}{3} \geq 2$$

$$n - \frac{m}{3} \geq 2$$

$$\rightarrow \frac{3n - m}{3} \geq 2$$

$$\text{or, } 3n - m \geq 6$$

$$m \leq 3n - 6 \quad \left[ \frac{dn}{n} < 6 \right]$$

$$2m \leq 6n - 12 \quad (x2)$$

$$\sum d(v) \leq 6n - 12$$

$$\frac{dn}{n} \leq \frac{6n - 12}{n}$$

$$\text{or, } \frac{dn}{n} \leq \frac{6n - 12}{n} \leq 6$$

$$\text{or, } \frac{dn}{n} \leq 6 - \frac{12}{n} < 6$$

$$\text{Avg degree } \left( \frac{dn}{n} \right) < 6$$

$\rightarrow$  Graph is 6 colorable

Q) Let  $G$  be a simple graph with maximum degree  $\Delta$ . Then  $\Delta \leq \chi'(G) \leq \Delta + 1$

If  $G$  is simple then either  $\chi'(G) = \Delta$

or  $\chi'(G) \leq \Delta + 1$

Contradict  $\rightarrow \chi'(G) > \Delta + 1$

Let  $\phi = (\phi_1, \phi_2, \dots, \phi_{\Delta+1})$  be an optimal

$(\Delta + 1)$  edge coloring of  $G$

So, prove -

$$\Delta \leq \chi'(G)$$

Let vertex  $v$  with degree  $\Delta$   
 $v$  has  $\Delta$  incident edges

at least  $\Delta$  colors

$$\chi'(G) \geq \Delta$$

And hence

is done

$$\text{So, } \chi'(G) \leq \Delta + 1$$

If the graph has 1 edge

$$\chi'(G) = 1 \leq \Delta + 1 \quad [\text{True}] \quad (\text{Basis Step})$$

Induction Hypothesis

Every simple graph with fewer than  $m$  edges can be edge colored - using at most

$\Delta + 1$  colors

Let graph  $G$  have  $m$  edges

We choose an edge,  $e = uv$  and remove

it. So,  $G - e$  has fewer edges  $\rightarrow$  so by

induction it has proper edge coloring using at most  $\Delta + 1$  colors.

Now let's try to color edge  $e$ .

$u \rightarrow$  one color missing at  $u$  ( $\Delta - 1$  colored incident edges)

$v \rightarrow$  one color missing at  $v$

Let,

$\alpha$  = color missing at  $u$

$\beta$  = color "

If  $\alpha = \beta \rightarrow$  we assign the color at edge  $e$

If  $\alpha \neq \beta \rightarrow$  Maximal path  $(\alpha/\beta)$   $\xrightarrow{\text{color}}$  Alternating path  $(\alpha, \beta)$



1 common color becomes free

assigned to  $e$

So, in every case  $e$  can be colored

using one of  $\Delta + 1$  colors.

$$\chi(G) \leq \Delta + 1$$